

1. Points $P = (-3, 4)$ and $Q = (-5, 2)$. Determine the exact magnitude of vector $\overrightarrow{PQ}$.
2. $f \circ g(x) = f(g(x))$. If $f(x) = 5x + 6$ and $g(x) = 3x - 1$, then $f \circ g(x) = kx + w$. Determine the ordered pair (k, w) .
3. Determine all complex roots for the equation $x^2 + 40 = -9$.
4. x and y are real numbers such that $x^2 + 3xy + y^2 = 60$. Determine the maximum possible value for the product (xy) .
5. The largest real value of x such that $\sin 3x + \sin x = 0$ when solved over the interval $[0, 2\pi)$ is $x = k\pi$. Determine the value of k . Express your answer as a common or improper fraction reduced to lowest terms.
6. Consider the graph of the function $y = \frac{1}{2}\cos(8\pi x - 4) + 12$. Let k represent the amplitude of this function and w represent its period. Determine the exact value of the product (kw) . Express your answer as a common or improper fraction reduced to lowest terms.
7. A sequence is defined by the rule $F_{n+1} = F_n + F_{n-1}$. $F_4 = 7$ and $F_7 = 29$. Determine the value of F_{10} .
8. Solve for all values of x such that $\log_{100}(x+2) = \log x$.